



# CARGO CONVOY CRM

Comprehensive User Manual & Technical Documentation

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## **Cargo Convoy Inc**

7119 Pennsylvania Ave, Upper Darby, PA, USA 19082

Phone: 267-513-0420 | MC: 01512751 | DOT: 4014885

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Built with Laravel 12 | PHP 8.4

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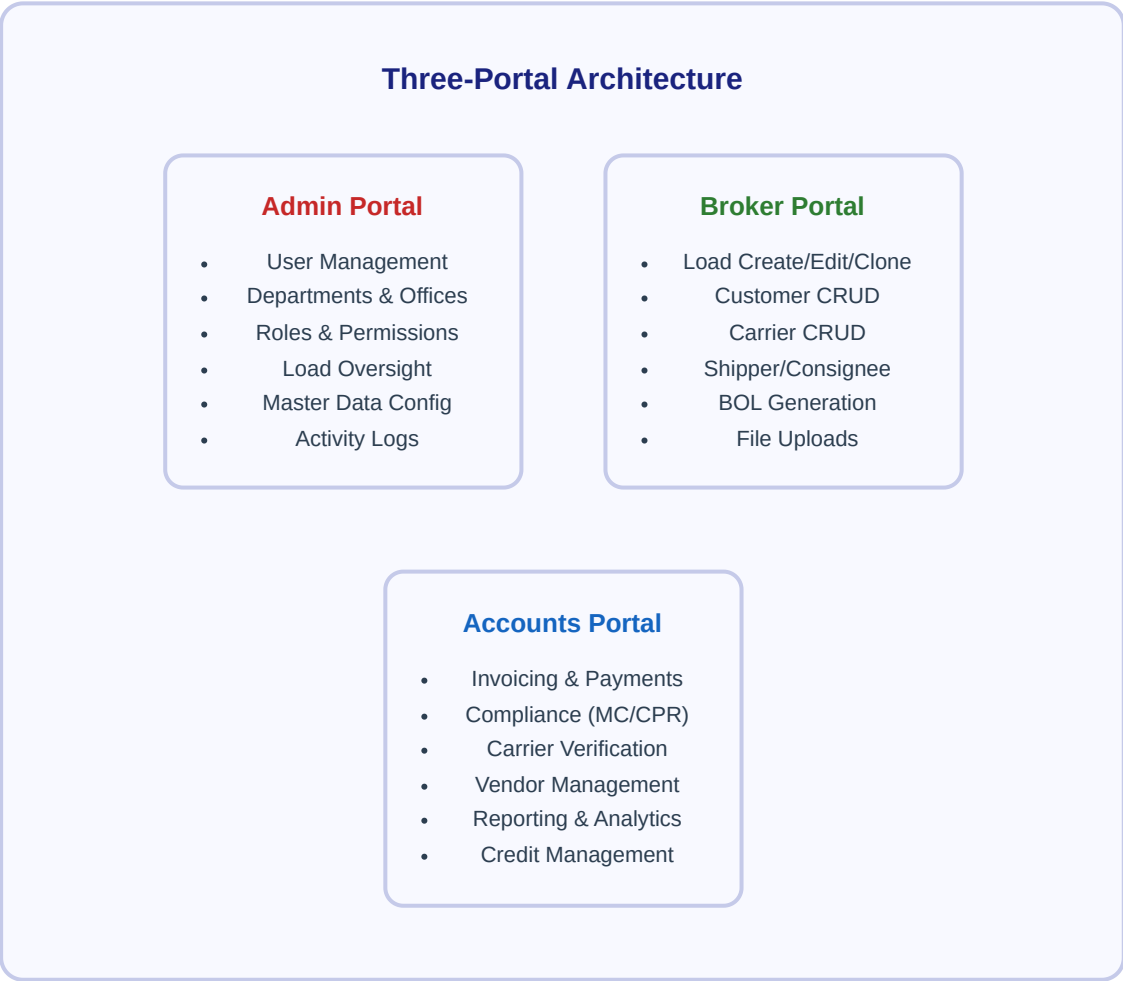
# 1. System Overview & Architecture

## 1.1 What is Cargo Convoy CRM?

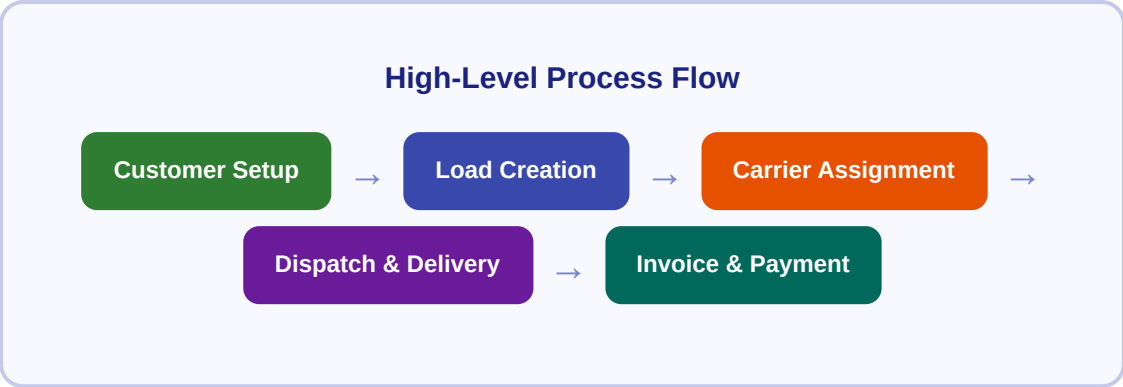
Cargo Convoy CRM is a comprehensive freight brokerage management system designed for **Cargo Convoy Inc.** It manages the complete lifecycle of freight loads from creation to payment, including customer relationships, carrier management, invoicing, compliance checks, and document handling.

**Technology Stack:** Laravel 11 (PHP 8.4) | MySQL Database | Blade Templates | Bootstrap 5 | jQuery | Dompdf for PDF generation | PhpSpreadsheet for Excel exports

## 1.2 System Architecture Diagram

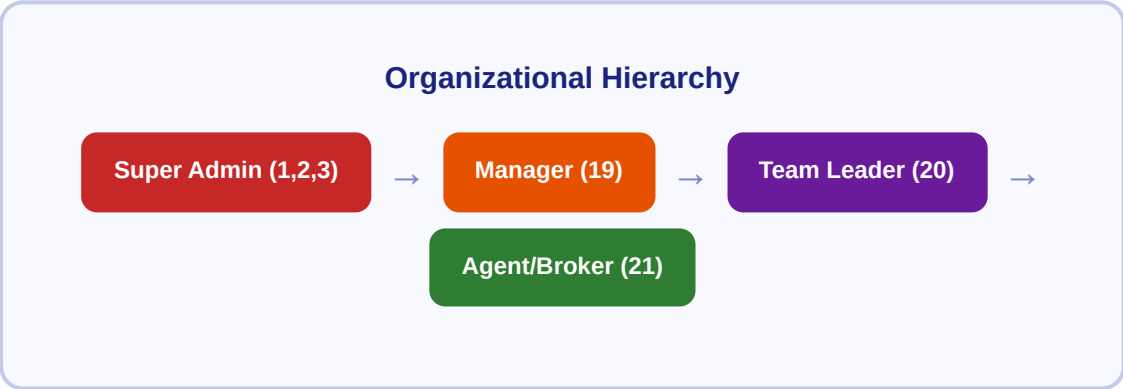


## 1.3 Application Flow Overview



## 2. User Roles & Access Control

### 2.1 Role Hierarchy



### 2.2 Role Access Matrix

| Role Group  | Role IDs    | Portal Access            | Landing Page           |
|-------------|-------------|--------------------------|------------------------|
| Admin       | 1, 2, 3, 22 | Admin + Broker + Account | /admin/admin_home      |
| Accounting  | 7, 8, 9, 24 | Account + Broker         | /account/accounting    |
| Vendor/AP   | 4, 5, 6     | Account + Broker         | /account/vendor-system |
| Compliance  | 10, 11, 12  | Account + Broker         | /account/compliance    |
| Reporting   | 13-18       | Account + Broker         | /account/reporting     |
| Manager     | 19          | Broker                   | /broker/load           |
| Team Leader | 20          | Broker                   | /broker/load           |

## 2.3 Middleware Access Control

```
// EnsureUserIsAuthenticated.php - URL Segment Permission
if (in_array($role_id, $accounts)) {
    $urls = ['account', 'broker', 'file', 'files', 'remaing'];
} elseif (in_array($role_id, $broker)) {
    $urls = ['broker', 'file', 'files', 'remaing'];
} elseif (in_array($role_id, $admin)) {
    $urls = ['admin', 'broker', 'account', 'file', 'files', 'remaing'];
}
// If user accesses unauthorized URL segment -> 404
```

## 2.4 Data Visibility Rules

**Admins (Roles 1,2,3):** See ALL data across all users

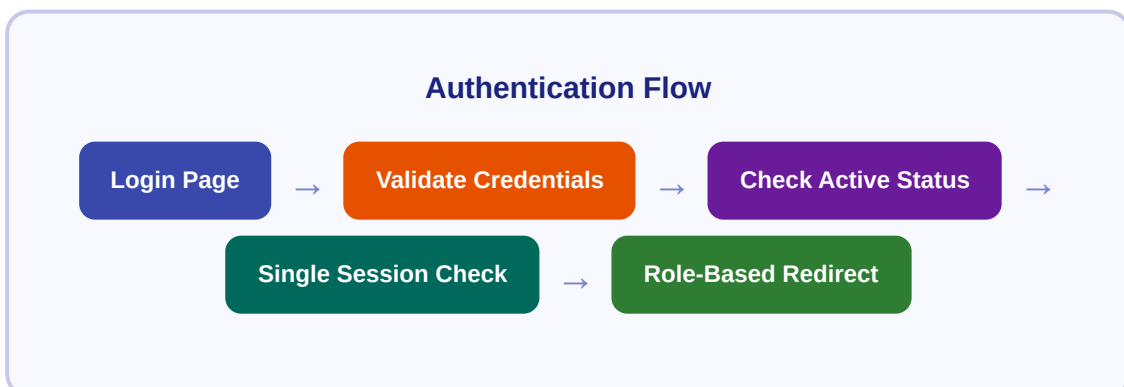
**Managers (Role 19):** See data of team leaders under them + agents under those TLs

**Team Leaders (Role 20):** See data of agents under them

**Agents (Role 21):** See only their own data

# 3. Authentication & Login Flow

## 3.1 Login Process Flowchart



## 3.2 Key Security Features

- 1 Single Session Enforcement:** Only one active session per user. If user is already logged in from another device, login is blocked (unless session expired > 1 minute).

- 2 **Active Status Check:** Only users with status = "active" can login. Inactive accounts are blocked.
- 3 **Session Management:** Old sessions are cleared on new login. LogoutOtherDevices is enforced.
- 4 **Activity Logging:** Every login/logout is recorded with username, IP, and timestamp.

### 3.3 Password Management

Users can change their password from the Profile section. Admins can reset passwords for any user via the admin panel.

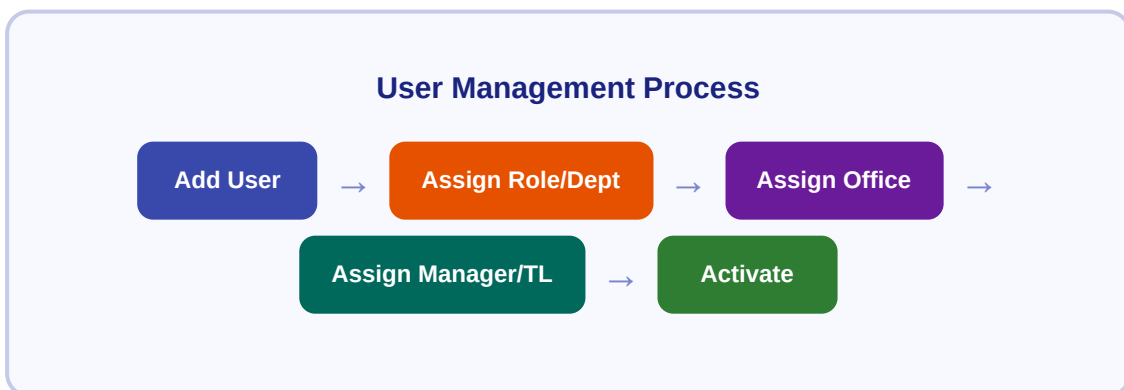
```
// Password change requires current password verification
Route::post('broker/change-password', [AdminController::class,
'update_password']);
Route::post('admin/admin-update-password/{id}', [AdminController::class,
'admin_update_password']);
```

## 4. Admin Panel Module

### 4.1 Admin Dashboard

The admin dashboard provides a comprehensive overview of all loads across the system with tabs for different statuses. Admins can filter by Office, Manager, Team Leader, and Agent.

### 4.2 User Management



| Feature     | Description                                                        | Route                  |
|-------------|--------------------------------------------------------------------|------------------------|
| Create User | Add new user with role, department, office, manager, TL assignment | POST admin/create-user |

|                |                                                     |                                       |
|----------------|-----------------------------------------------------|---------------------------------------|
| Update Status  | Activate/Deactivate user accounts                   | POST admin/update-status              |
| Reset Password | Admin can reset any user's password                 | POST admin/admin-update-password/{id} |
| View Users     | Separate views for Broker, Account, and Admin users | GET admin/broker-users                |

## 4.3 Organization Structure Management

| Entity       | Operations | Purpose                                           |
|--------------|------------|---------------------------------------------------|
| Departments  | CRUD       | Broker (3), Account (2), Admin (1), IT (6,7)      |
| Offices      | CRUD       | Physical office locations                         |
| Managers     | CRUD       | Link users as managers within offices/departments |
| Team Leaders | CRUD       | Link users as TLs under managers                  |

## 4.4 Master Data Configuration

| Master Data        | Examples                          |
|--------------------|-----------------------------------|
| Equipment Types    | Dry Van, Flatbed, Reefer, etc.    |
| Shipment Types     | FTL, LTL, TONU, etc.              |
| Status Types       | Open, Delivered, Completed, etc.  |
| Countries & States | Geographic data for address forms |
| IP Configuration   | IP whitelist for security         |

## 4.5 All Data View

Admins have access to a consolidated "All Data" view with tabs for Customers, Carriers, Shippers, Consignees, and Loads with full search capability.

# 5. Broker Module

## 5.1 Broker Dashboard

The Broker portal is the primary workspace for freight brokers/agents. After login, brokers land on the Load Management page where they can manage their loads through different status tabs.

### 5.2 Broker Module Features

**Load Management**

- Create New Load
- Edit Load
- Clone Load
- Change Status
- Search & Filter

**Customer Management**

- Add Customer
- Edit Customer
- Credit Monitoring
- Remittance Upload

**Carrier Management**

- My Carriers
- All Carriers
- Add/Edit Carrier
- MC/DOT Search

**Shipper/Consignee**

- Add Shipper
- Add Consignee
- Cross-reference
- Auto-fill in Loads

### 5.3 Navigation Structure

Broker Portal URLs:

/broker/load

- Load listing (main page)

/broker/load-create

- Create new load

/broker/edit-load/{id}

- Edit existing load

/broker/clone/load/{id}

- Clone existing load

/broker/customer

- Customer management

/broker/carrier

- Carrier management

/broker/shipper

- Shipper management

/broker/Consignee

- Consignee management

/broker/profile

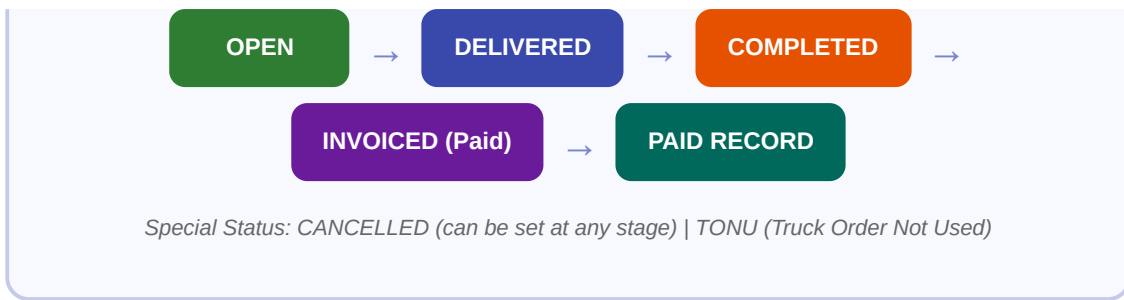
- User profile settings

## 6. Load Management (Core Module)

### 6.1 Load Lifecycle Flowchart

**Load Status Workflow**





## 6.2 Load Creation Process

- 1 **Select Customer (Bill To):** Choose an approved customer. System validates credit limit before allowing creation.
- 2 **Set Load Type:** OTR (Over The Road) or DRAYAGE. Type determines rate field editability.
- 3 **Add Shipper(s):** Multiple shippers supported. Select from saved shippers or enter new. Includes location, appointment date/time, commodity details.
- 4 **Add Consignee(s):** Multiple consignees with delivery appointments. Delivery date must be after pickup date.
- 5 **Assign Carrier:** Search by name, MC#, or DOT#. Auto-fills carrier details. System validates carrier exists in database.
- 6 **Set Rates:** Customer rate + FSC + other charges = Final Rate. Carrier fee must not exceed customer final rate.
- 7 **Credit Validation:** System checks if new load amount would exceed customer's assigned credit limit.

## 6.3 Credit Limit Validation

```

// Credit limit check during load creation
$assignedCreditLimit = $customer->adv_customer_credit_limit;
$totalLoadCreationAmount = Load::where('customer_id', $customerId)
    ->where('load_status', '!=', 'Cancelled')
    ->sum('shipper_load_final_rate');
$newTotal = $totalLoadCreationAmount + $loadAmount;

if ($newTotal > $assignedCreditLimit) {
    return error("Cannot create load. Available credit exceeded.");
}

```

## 6.4 Load Tabs & Filtering

| Tab | Filter Criteria | Description |
|-----|-----------------|-------------|
|-----|-----------------|-------------|

|           |                                          |                                     |
|-----------|------------------------------------------|-------------------------------------|
| All Loads | No filter                                | Shows all loads (paginated 50/page) |
| Open      | load_status = 'Open'                     | Active loads awaiting dispatch      |
| Delivered | load_status = 'Delivered'                | Loads confirmed delivered           |
| Complete  | load_status = 'Completed' AND no invoice | Ready for invoicing                 |
| Invoiced  | invoice_status = 'Paid'                  | Invoice generated/sent              |
| Paid      | invoice_status = 'Paid Record'           | Payment received                    |

## 6.5 Load Clone Feature

Brokers can clone an existing load to quickly create similar shipments. The clone copies all shipper/consignee info, customer, carrier, and other details but resets rates to 0 and status to "Open".

## 6.6 Rate Calculation Logic

### Rate Calculation Formula

#### Customer Side:

Customer Rate + (FSC% x Customer Rate) + Other Charges = **Shipper Final Rate**

#### Carrier Side:

Carrier Fee + (Billing FSC% x Carrier Fee) + Other Charges = **Final Carrier Fee**

**Profit Margin:** Shipper Final Rate - Final Carrier Fee = **Margin**

# 7. Customer Management

## 7.1 Customer Lifecycle

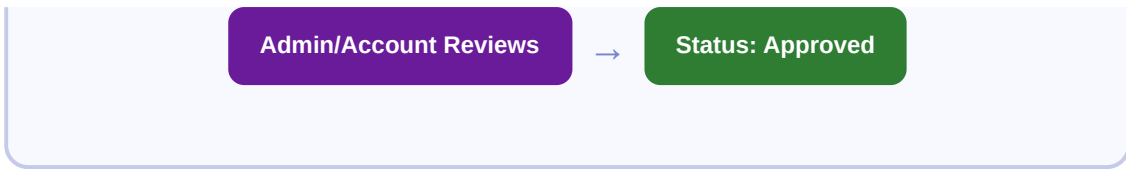
### Customer Approval Workflow

Broker Creates Customer



Status: Not Approved





**Important:** Only approved customers can be selected when creating loads. The system enforces this validation at both form and controller level.

## 7.2 Customer Data Fields

| Section         | Fields                                                              |
|-----------------|---------------------------------------------------------------------|
| Basic Info      | Name, MC/FF Type & Number, Corporation Type, Status                 |
| Address         | Street, Country, State, City, ZIP                                   |
| Billing Address | Separate billing address fields                                     |
| Contacts        | Primary contact, telephone, email, secondary contact, billing email |
| Advanced        | Currency, Credit Limit, Payment Terms, Factoring Company, Website   |
| Documents       | File uploads (multiple), Remittance files                           |

## 7.3 Credit Limit System

```
// Credit Summary Calculation (helpers.php)
function calculate_customer_credit_summary($customer, $assignedCreditLimit,
$loadCreateAmount, $receivingAmount) {
    $usedAmount      = max(0, $loadCreateAmount - $receivingAmount);
    $remainingCredit = max(0, $assignedCreditLimit - $loadCreateAmount);
    return ['assigned_credit_limit', 'used_amount', 'remaining_credit'];
}
```

## 7.4 Customer Approval Form

Brokers can submit a Customer Approval Form which goes through the admin/account team for review. Once approved, the credit limit is assigned and the customer becomes available for load creation.

## 7.5 Remittance Management

Both Brokers and Account teams can upload remittance files (PDFs, images) against customers. Files are tagged with timestamps and notes, and can be filtered by date range.

## 8. Carrier Management

### 8.1 Carrier Data Structure

#### Carrier (External) Entity

**Identification:** Carrier Name, MC/FF Number, DOT Number  
**Address:** Country, State, City, ZIP  
**Contact:** Contact Name, Email, Telephone, Extension, Fax  
**Business:** Payment Terms, Factoring Company, Load Type, Corporation  
**Status:** Active/Blacklisted/Blocked  
**Documents:** File uploads (W9, Insurance, Authority Letter)

### 8.2 Carrier Views

| View         | Description                            | Access            |
|--------------|----------------------------------------|-------------------|
| My Carriers  | Carriers added by the logged-in broker | Agent's own       |
| All Carriers | All carriers in the system             | Searchable by all |

### 8.3 Carrier Validation

**Unique MC#:** System enforces unique MC numbers. Duplicate MC# is rejected.  
**Unique DOT#:** DOT numbers must also be unique across all carriers.  
**Load Assignment:** When assigning a carrier to a load, the system verifies the carrier exists in the database.

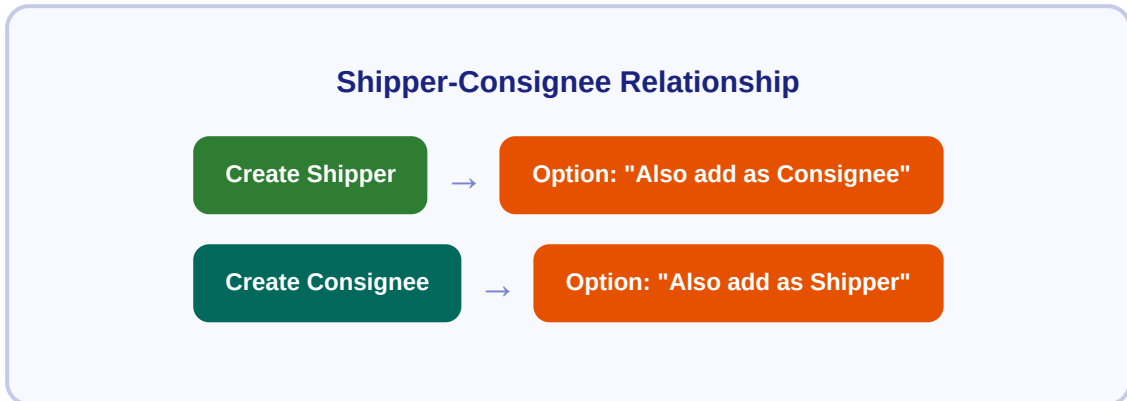
### 8.4 Carrier Auto-Suggest in Load Form

When typing carrier name, MC#, or DOT# in the load form, the system provides live suggestions via AJAX. Selecting a suggestion auto-fills all carrier fields.

```
// Carrier suggestion endpoint
Route::post('broker/fetch-carrier-suggestions', [LoadController::class,
'fetchCarrierSuggestions']);
// Returns: id, carrier_name, mcNumber, dotNumber, phone
```

## 9. Shipper & Consignee Management

### 9.1 Shipper/Consignee Cross-Reference



### 9.2 Data Fields

| Field      | Shipper                               | Consignee                          |
|------------|---------------------------------------|------------------------------------|
| Name       | shipper_name                          | consignee_name                     |
| Address    | Full address (country/state/city/zip) | Full address                       |
| Contact    | Name, email, phone, ext, fax          | Name, email, phone, ext, fax       |
| Operations | Hours, appointments, intersections    | Hours, appointments, intersections |
| Notes      | Shipping notes, internal notes        | Shipping notes, internal notes     |

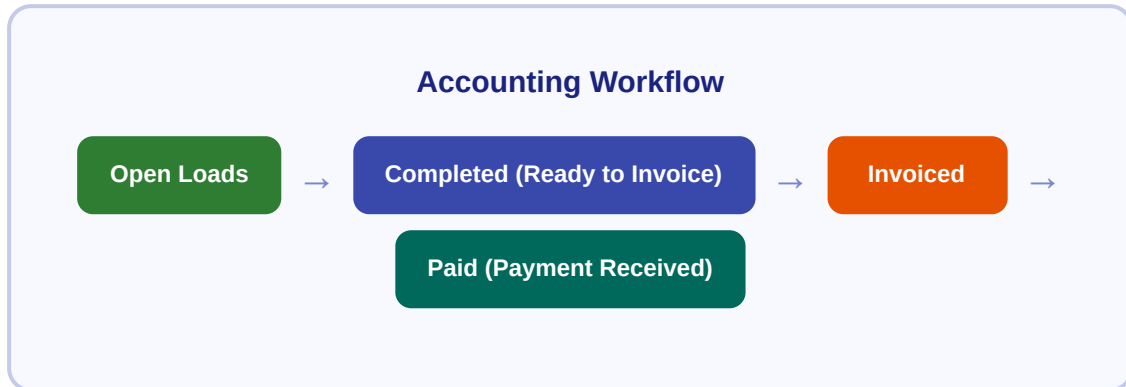
### 9.3 Usage in Load Creation

When creating or editing a load, brokers can select from their saved shippers and consignees. The system auto-fills the address fields. Multiple shipper/consignee stops are supported (up to 15).

```
// Auto-fill shipper details in load form
Route::get('broker/fetch-shipper-details', [LoadController::class,
'fetchShipperDetails']);
Route::get('broker/fetch-consignee-details', [LoadController::class,
'fetchConsigneeDetails']);
```

## 10. Accounting & Invoicing

### 10.1 Accounting Dashboard Tabs



### 10.2 Invoice Generation

- 1 **Mark as Invoiced:** Account team changes invoice\_status to 'Paid' (meaning invoiced to customer)
- 2 **Generate Invoice PDF:** System generates invoice with load details, customer info, rates
- 3 **Email Invoice:** Send invoice via email with attachments (merged PDF of all documents)
- 4 **Record Payment:** Update receiving\_amount when payment is received
- 5 **Mark as Paid Record:** Final status when fully paid

### 10.3 Multi-Load Invoicing

The system supports generating a single invoice for multiple loads (PI - Proforma Invoice). Account team enters comma-separated load numbers to generate a combined invoice.

```
// Multi-load invoice generation
Route::Post('account/loads-pi', [AccountController::class, 'loadspi']);
Route::Post('account/loads-multipal-invoice', [AccountController::class, 'loadsMultipalInvoice']);
```

### 10.4 Carrier Payment Management

| Feature   | Description                        |
|-----------|------------------------------------|
| Quick Pay | Mark carrier for expedited payment |

|                  |                                        |
|------------------|----------------------------------------|
| Payment Method   | Set payment method (ACH, Check, Zelle) |
| Ready to Pay     | Flag load as ready for carrier payment |
| Carrier Due Date | Set carrier payment due date           |
| Mark as Paid     | Record carrier payment completion      |
| Upload Documents | Attach carrier invoices and receipts   |

## 10.5 Vendor Management System

The Vendor System provides a consolidated view of all loads with carrier payment status, internal notes, and quick actions for the accounts payable team.

## 10.6 AR Aging Close (New)

**New Feature:** Account team can now close aging records directly from the Paid tab when payments are fully reconciled.

- 1 View Aging:** See aging days (days since invoice) for each paid load
- 2 Review Payment:** Check excess/remaining amounts after partial payments
- 3 Close Aging:** Click "Close" to mark the aging record as reconciled

Route: POST /account/update-ar-aging-close

## 10.7 All Load Status AR View (New)

A comprehensive dashboard showing **all loads across all statuses** for the AR team. Includes:

| Feature          | Description                                                     |
|------------------|-----------------------------------------------------------------|
| All Status Tabs  | View Open, Delivered, Completed, Invoiced, Paid all in one page |
| Advanced Filters | Filter by office, agent, date range, customer                   |
| Excess/Remaining | Track overpayment and underpayment per load                     |
| Bulk Actions     | Process multiple loads at once                                  |

Route: GET /account/all-status-loads

## 10.8 Custom Invoice Subject (New)

Account team can now customize the email subject line when sending invoices. The default format is:

```
Invoice For Load #{load_no} (#{invoice_no}) REF #{reference_no}
```

Users can edit this before sending, or leave it as the default.

## 10.9 Credit Limit Log Export (New)

Export the complete credit limit change history for all customers as an Excel spreadsheet. Useful for auditing credit changes over time.

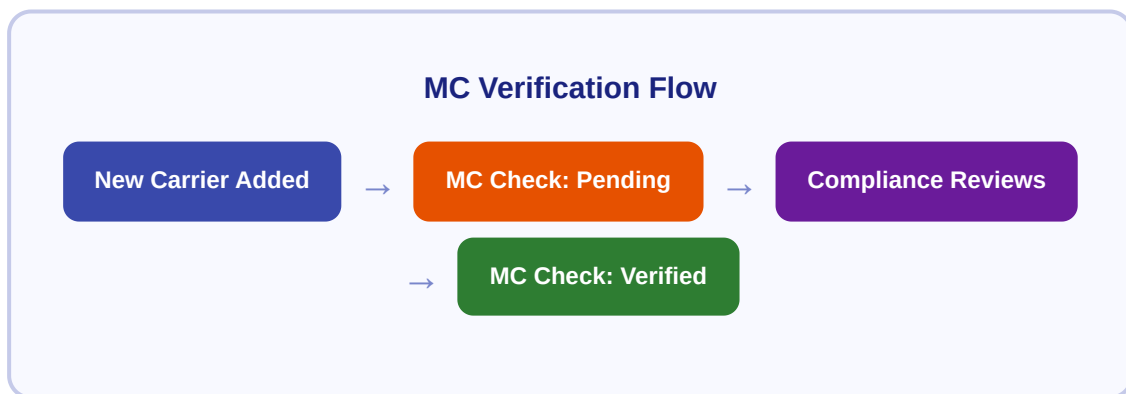
Route: GET /account/customers/export-credit-limit-log

# 11. Compliance Module

## 11.1 Compliance Overview

The compliance module ensures all carriers meet regulatory requirements before loads can be dispatched. It covers MC (Motor Carrier) checks and CPR (Carrier Performance Review) checks.

## 11.2 MC Check Process



## 11.3 CPR (Carrier Performance Review) Check

Each load can have a CPR status that indicates whether the carrier assignment has been reviewed and approved by compliance.

| Status       | Meaning                                       |
|--------------|-----------------------------------------------|
| Not Approved | Default - carrier assignment not yet reviewed |



|              |                                      |
|--------------|--------------------------------------|
| Verified     | Carrier approved for this load       |
| Not Verified | Carrier rejected - needs replacement |

## 11.4 Carrier Verification

Detailed carrier verification includes factoring company verification, bank document uploads, and insurance validation.

```
Route::get('account/carrier-verification', [AccountController::class,
'carrier_verification']);
Route::post('account/carrier-verification/save', [AccountController::class,
'carrier_verification_save']);
```

## 11.5 Carrier Block/Unblock

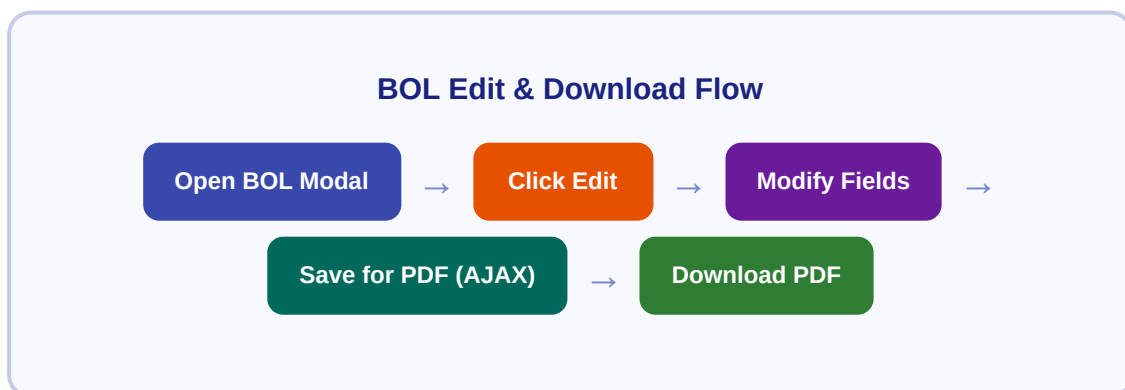
Compliance team can block carriers who fail compliance checks. Blocked carriers cannot be assigned to new loads.

# 12. Bill of Lading (BOL)

## 12.1 BOL Feature Overview

The Bill of Lading module allows brokers to generate, edit, and download BOL documents as PDFs. Edited data is saved and persists for future downloads.

## 12.2 BOL Workflow



## 12.3 BOL Document Sections

| Section            | Content                                            |
|--------------------|----------------------------------------------------|
| Header             | Company logo, name, address, phone                 |
| Load Info          | Load Number, BOL Number, Ship Date, Delivery Date  |
| Shipper            | Shipper name(s) and address(es)                    |
| Consignee          | Consignee name(s) and address(es)                  |
| 3rd Party Billing  | Third-party billing information                    |
| Transportation Co. | MC#, Carrier Name                                  |
| Freight Table      | Pieces, Description, Weight, Type, NMFC, HM, Class |
| Notes & COD        | Notes, COD Amount, COD Fee, Declared Value         |
| Signatures         | Shipper, Carrier, Consignee signatures with dates  |

## 12.4 Technical Implementation

```
// Save BOL edit data (AJAX POST)
Route::post('/broker/load/{id}/bol/save', [LoadController::class,
'saveBolEditData']);

// Download BOL PDF (GET - reads from saved bol_edit_data)
Route::get('/broker/load/{id}/bol/pdf', [LoadController::class,
'generateBolPdf']);

// PDF generated using Dompdf library with 'letter' paper size
```

# 13. Reporting & Analytics

## 13.1 Report Types



[Credit/Limit Report](#)[Load Completed Log](#)

## 13.2 Report Details

| Report              | Key Metrics                                                               | Export |
|---------------------|---------------------------------------------------------------------------|--------|
| Carrier Report      | Total revenue per carrier, load count, open/delivered/completed breakdown | Excel  |
| Customer Report     | Revenue per customer, margin, credit limit usage                          | Excel  |
| Customer Detail     | Full customer listing with all details                                    | Excel  |
| Dispatcher Report   | Revenue per agent, carrier fees, margins, load counts                     | Excel  |
| Load Report         | Individual load details with user info                                    | Excel  |
| Sales Rep Report    | Shipper rates vs carrier fees per agent                                   | Excel  |
| Aging Report        | Invoice aging (days since invoice date)                                   | Excel  |
| Credit/Limit Report | Customer credit limits and remaining credit                               | Excel  |

## 13.3 Sales Dashboard Chart

The reporting module includes an interactive chart showing daily revenue trends with filters for:

- **Date Range:** Custom start and end dates (default: last 7 days)
- **Office:** Filter by office location
- **Agent:** Filter by specific agent

Chart displays: Shipper Rates (revenue), Carrier Fees (cost), and Margin (profit) by date.

## 13.4 Export Functionality

All reports support Excel export using PhpSpreadsheet library. PDF exports are available for specific reports using Dompdf.

```
// Example: Excel export routes
Route::get('account/CarrierReportingExcel', [AccountController::class,
'CarrierReportingExcel']);
Route::get('account/customerReportingExcel', [AccountController::class,
'customerReportingExcel']);
```

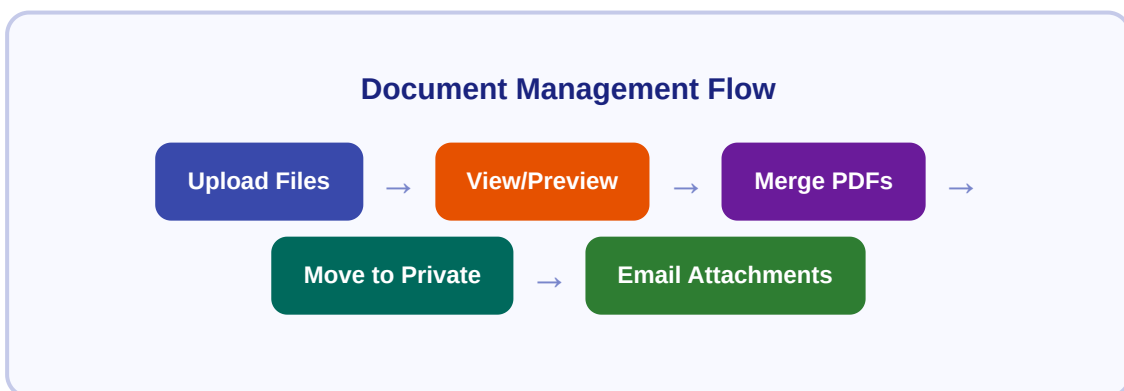
```
Route::get('account/dispatcherReportingExcel', [AccountController::class, 'dispatcherReportingExcel']);
Route::get('account/agingReportingExcel', [AccountController::class, 'agingReportingExcel']);
```

## 14. Document Management & File Uploads

### 14.1 Load Document Types

| Document Type             | Field Name                | Purpose                             |
|---------------------------|---------------------------|-------------------------------------|
| Carrier Rate Confirmation | carrer_rate_cnfrm_doc     | Signed rate agreement with carrier  |
| Proof of Delivery (POD)   | pod_doc                   | Delivery confirmation document      |
| Shipper Rate Approval     | shipper_rate_approval_doc | Customer rate approval              |
| Carrier Invoice           | carrier_invoice_doc       | Carrier's invoice for payment       |
| Delivery Order (DO)       | do                        | Original delivery order document    |
| Optional Documents        | optional_docs             | Any additional supporting documents |

### 14.2 File Operations



### 14.3 PDF Merge Feature

Multiple uploaded documents (PDFs and images) can be merged into a single PDF per load. This merged file is used for emailing invoices with all supporting documents.

```
// Merge files endpoint
Route::post('files/merge-files', [FilesUploadController::class,
'mergeFiles']);
// Uses TCPDF + FPDF for PDF merging
// Images (jpg, png, gif) are converted to PDF before merging
```

## 14.4 File Storage Structure

```
public/uploads/
|-- Load-files/{load_number}-{customer_id}/    (Public load documents)
|-- delivery-order/{load_id}/                  (Private delivery orders)
|-- carrier/                                    (Carrier documents)
|-- customers/{customer_name}/                 (Customer files)
|-- remittances/                               (Remittance files)
|-- mergedfiles/                               (Merged PDFs)
```

# 15. Email System

---

## 15.1 Invoice Email

The system sends invoice emails to customers with all relevant documents attached. Multiple recipients (TO and CC) are supported.

## 15.2 Email Process

- 1 **Select Documents:** Choose which load documents to attach (Rate Con, POD, Invoice, etc.)
- 2 **Set Recipients:** Enter email addresses (comma-separated for multiple). Optional CC field.
- 3 **Auto-Merge:** If multiple files selected, system merges them into one PDF attachment.
- 4 **Send:** Email sent via SMTP with professional template including load#, invoice#, ref#.

## 15.3 Email Template

```
Subject: Invoice For Load #{load_no} (#{invoice_no}) REF #{reference_no}
From: ar@cargoconvoy.co (Cargoconvoy)
Reply-To: ar@cargoconvoy.co

Body:
"Greetings!
```

Kindly find the attached invoice for load #{load\_no} ({invoice\_no}) Ref # {ref\_no}

Thanks & Regards

{Username}

Accounts Receivable

Cargo Convoy Inc

Mailing Address: 7119, Pennsylvania Ave, Upper Darby, PA - 19082

MC - 01512751 | DOT - 4014885"

## 15.4 Security

**Attachment Security:** Only files from authorized directories (uploads/delivery-order and uploads/) can be attached. The system validates file paths to prevent arbitrary file access.

# 16. Database Schema

## 16.1 Core Tables

| Table                | Purpose                  | Key Relationships                                          |
|----------------------|--------------------------|------------------------------------------------------------|
| users                | All system users         | belongs_to: role, department, office, manager, team_leader |
| loads                | Core freight loads       | belongs_to: user, customer, carrier(external)              |
| customers            | Customer entities        | belongs_to: user; has_many: loads                          |
| externals            | Carriers                 | belongs_to: user; has_many: loads                          |
| shippers             | Shipper locations        | belongs_to: user                                           |
| consignees           | Consignee locations      | belongs_to: user                                           |
| roles                | User roles               | has_many: role_has_permissions                             |
| permissions          | System permissions       | belongs_to: department                                     |
| role_has_permissions | Role-permission mapping  | belongs_to: role, permission                               |
| departments          | Organization departments | has_many: users                                            |

|                      |                         |                                      |
|----------------------|-------------------------|--------------------------------------|
| offices              | Office locations        | has_many: users                      |
| mangers              | Manager assignments     | belongs_to: user, office, department |
| team_leaders         | Team leader assignments | belongs_to: manager                  |
| logs                 | Activity audit trail    | records all CRUD operations          |
| carrier_verification | Carrier compliance data | belongs_to: load                     |
| load_payments        | Payment records         | belongs_to: load                     |

## 16.2 Loads Table - Key Columns

| Column Group     | Fields                                                              |
|------------------|---------------------------------------------------------------------|
| Identity         | id, load_number, user_id, customer_id, carrier_id                   |
| Status           | load_status, invoice_status, cpr_check                              |
| Customer Rates   | load_shipper_rate, load_fsc_rate, shipper_load_final_rate           |
| Carrier Rates    | load_carrier_fee, load_billing_fsc_rate, load_final_carrier_fee     |
| Shipper (JSON)   | load_shipperr, load_shipper_location, load_shipper_appointment      |
| Consignee (JSON) | load_consignee, load_consignee_location, load_consignee_appointment |
| Invoice          | invoice_number, invoice_date, invoice_status, receiving_amount      |
| Documents        | public_file, load_delivery_do_file, carrierDoc                      |
| BOL              | bol_edit_data (JSON - saved BOL edits)                              |

# 17. API Routes & Endpoints

## 17.1 Admin Routes

| Method | Endpoint          | Purpose                        |
|--------|-------------------|--------------------------------|
| GET    | /admin/admin_home | Admin dashboard with all loads |

|      |                      |                          |
|------|----------------------|--------------------------|
| GET  | /admin/broker-users  | List broker users        |
| POST | /admin/create-user   | Create new user          |
| POST | /admin/update-status | Activate/deactivate user |
| GET  | /admin/department    | Department management    |
| GET  | /admin/office        | Office management        |
| GET  | /admin/manager       | Manager management       |
| GET  | /admin/team_leader   | Team leader management   |
| GET  | /admin/roles         | Roles & permissions      |
| GET  | /admin/all_data      | Consolidated data view   |
| GET  | /admin/activity_logs | System activity logs     |

## 17.2 Broker Routes

| Method | Endpoint                   | Purpose                |
|--------|----------------------------|------------------------|
| GET    | /broker/load               | Load listing with tabs |
| POST   | /broker/load-create        | Create new load        |
| GET    | /broker/edit-load/{id}     | Edit load form         |
| POST   | /broker/load/update/{id}   | Update load            |
| GET    | /broker/clone/load/{id}    | Clone load             |
| GET    | /broker/load/{id}/bol/pdf  | Download BOL PDF       |
| POST   | /broker/load/{id}/bol/save | Save BOL edit data     |
| GET    | /broker/customer           | Customer list          |
| POST   | /broker/customer/create    | Create customer        |
| GET    | /broker/carrier            | Carrier list           |
| POST   | /broker/create/carrier     | Create carrier         |
| GET    | /broker/shipper            | Shipper list           |
| GET    | /broker/Consignee          | Consignee list         |



## 17.3 Account Routes

| Method | Endpoint                                   | Purpose                          |
|--------|--------------------------------------------|----------------------------------|
| GET    | /account/accounting                        | Accounting dashboard             |
| GET    | /account/compliance                        | Compliance dashboard             |
| GET    | /account/reporting                         | Reports & analytics              |
| GET    | /account/vendor-system                     | Vendor management                |
| GET    | /account/carrier-verification              | Carrier verification             |
| POST   | /account/mc-check                          | MC check update                  |
| POST   | /account/cpr-check                         | CPR check update                 |
| POST   | /account/update-invoice-status/{id}        | Mark as invoiced                 |
| POST   | /account/send-mail                         | Send invoice email               |
| GET    | /account/credit                            | Credit management                |
| POST   | /account/update-ar-aging-close             | Close AR aging record            |
| GET    | /account/all-status-loads                  | All Load Status AR view          |
| GET    | /account/customers/export-credit-limit-log | Export credit limit log to Excel |

## 17.4 Activity Logging

```
// Every significant action is logged via helpers.php
addToLog($customer_id, $load_id, $subject, $old_data, $new_data);

// Log record contains:
// - user_name, user_id, user_email
// - IP address, full URL
// - old_json (before state), new_json (after state)
// - message (human-readable description)
```

Commit: 23e369c - customer approval form mail functionality (2026-08-26 18:37:51 +0530)

Routes: 310 | Controllers: 15 | Models: 34 | Migrations: 35 | Views: 124